

Linear Algebra (MTS203)

Term-1 Exam, (120 mins — 20 Marks)

Class no. 7500

Exam Date - October 13, 2023

Question - 1 (5 marks)

Let $A\mathbf{x} = \mathbf{b}$ be a linear system whose augmented matrix $(A|B)$ has reduced row echelon form,

$$\left[\begin{array}{ccccc|c} 1 & 2 & 0 & 3 & 1 & -2 \\ 0 & 0 & 1 & 2 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

- a) Find all solutions to the system.
- b) If $\mathbf{a}_1 = [1, 1, 3, 4]^T$, and $\mathbf{a}_2 = [2, -1, 1, 3]^T$, determine $\mathbf{b}$.

Question - 2 (5 marks)

Let A be an $n \times n$ matrix and let

$$B = A + A^T \quad \text{and} \quad C = A - A^T$$

A matrix is *symmetric* if its transpose is equal to itself, and *skew-symmetric* if its transpose is equal to its negative. (a) Prove that B is symmetric, and C is skew symmetric. (b) Prove that every square matrix of order n can be written as a sum of a symmetric matrix, and a skew-symmetric matrix. (c) Validate Part (b) by giving an example of a square matrix of order 4×4 .

Question - 3 (5 marks)

If A and B are $n \times n$ matrices, then prove that $\det(AB) = \det(A)\det(B)$. Use elementary matrices to show this, furthermore, do not imply any results on elementary matrices, prove intermediate results on elementary matrices as and when required.

Question - 4 (5 marks)

Let

$$A = \begin{bmatrix} A_{11} & A_{12} \\ O & A_{22} \end{bmatrix}$$

where all four blocks are $n \times n$ matrices. If A_{11} , and A_{22} are nonsingular, show that A must also be nonsingular, and that A^{-1} must be of the form

$$\left[\begin{array}{c|c} A_{11}^{-1} & C \\ \hline O & A_{22}^{-1} \end{array} \right]$$

Determine C .

Question - 5 (5 marks)

Perform the following block multiplication:

$$\left[\begin{array}{cc|cc} \frac{3}{5} & -\frac{4}{5} & 0 & 0 \\ \frac{4}{5} & \frac{3}{5} & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \end{array} \right] \left[\begin{array}{cc|c} \frac{3}{5} & \frac{4}{5} & 0 \\ -\frac{4}{5} & \frac{3}{5} & 0 \\ \hline 0 & 0 & 1 \\ 0 & 0 & 0 \end{array} \right]$$